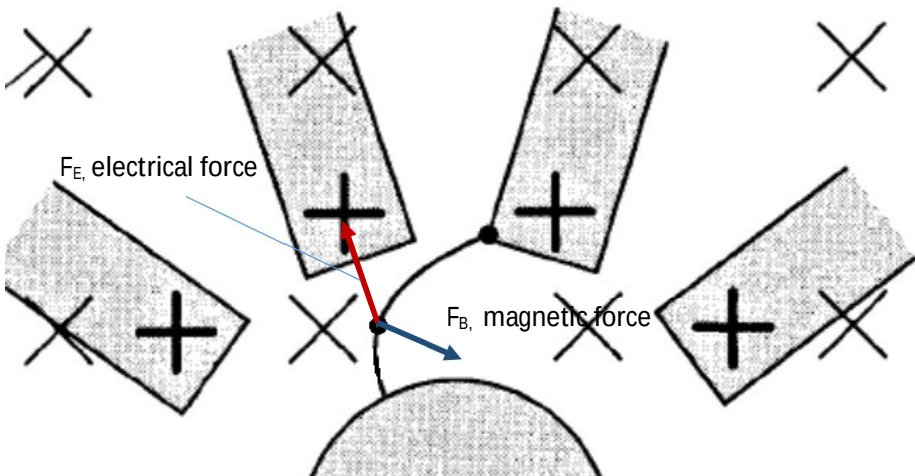
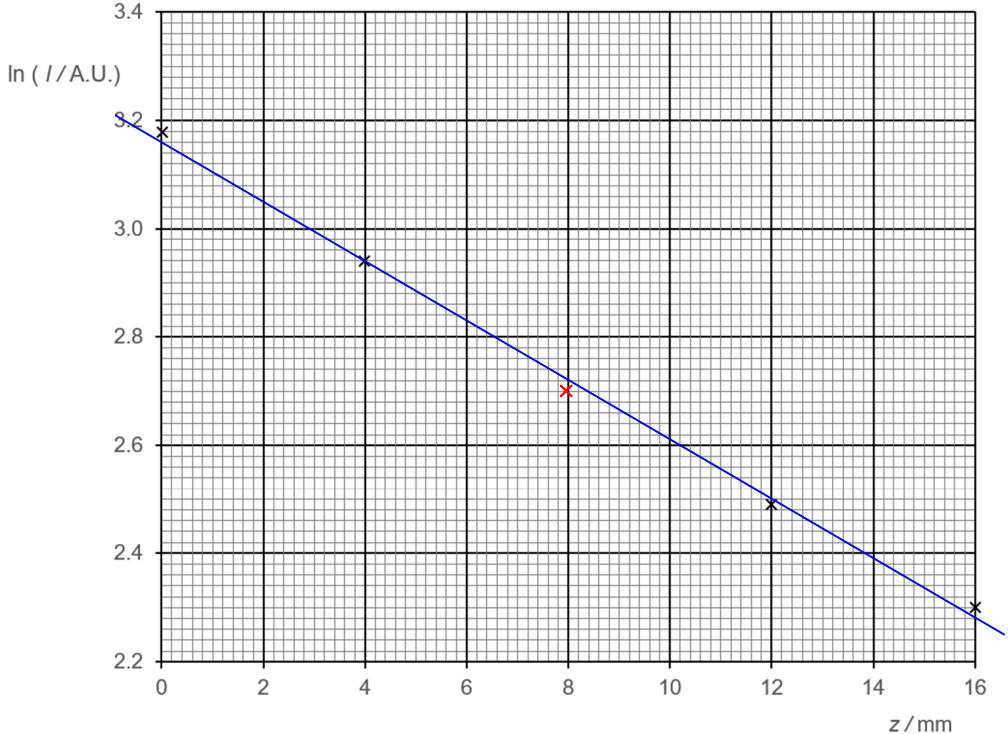


Question	Answer	Marks
7(a)(i)	To <u>prevent microwaves/(em) radiation</u> from <u>leaking/escaping out/exiting</u> of the cage/microwave oven.	1
(a)(ii)	From the passage, by international convention, wavelength of microwave oven used is given by 2.45 GHz. Wavelength of the microwave = $\frac{3.00 \times 10^8}{2.45 \times 10^9 \text{ Hz}} = 0.122 \text{ m}$ 100 times smaller than 0.122 m or 12.2 cm = $\frac{1.22 \times 10^{-3} \text{ m}}{100}$ or 0.0012 m. [M1] mark is awarded for using the information of 2.45 GHz for microwave oven. [A1] spacing < 1/100 of the wavelength of the microwave. First M1 mark must be correct before second mark is given. [-1] Powers of Ten error for conversion of GHz	M1 A1
(b)(i)	Potential difference across the electrodes = 5000 V By conservation of energy, <u>Kinetic energy gained = electrical potential energy loss</u> = $(5000)(1.6 \times 10^{-19}) = 8.0 \times 10^{-16} \text{ J}$	1
(b)(ii)	$P = (\text{Energy of the electron}) \times (n / t)$ Each electron has available max. $8.0 \times 10^{-16} \text{ J}$ energy (assuming that the electrons start off from cathode with negligible kinetic energy) to be converted to microwave. Least number of electrons per second $= \frac{1000 \text{ W}}{8.0 \times 10^{-16}} = 1.25 \times 10^{18}$ (Allow value calculated from (b)(i)) Alternative method: $P = IV \Rightarrow I = 1000 / (5.00 \times 10^3) = 0.200 \text{ A}$ $I = Q/t = (n/t) (1.60 \times 10^{-19}) = 0.200 \text{ A}$ $n/t = 0.200 / (1.60 \times 10^{-19}) = 1.25 \times 10^{18}$	1
b(iii)	<u>Not all the (kinetic) energy of the electrons is converted into the energy of the microwaves</u> as : - Electrons gives off e.m. radiation of varying wavelengths as it accelerates towards the anode and hence actual energy possessed by electrons are lower when reaching the anode. - some electrons hit the anode, some of its kinetic energy is also converted to heat energy / passed to the molecules (or atoms) in the anode causing thermal agitation/ converted to so less energy is available for conversion to microwave energy Not all the microwaves generated from the energy is fed into the cavity resulting in energy losses due to :	1

	- the walls in the cavity of the food chamber may absorb some of the microwave, - microwaves may be fed back / coupled back to the magnetron (resulting in actual useful power of microwave less than the actual energy that can be supplied by the electrons). Answer related must be relate back to the efficiency of <u>conversion of energy of the electrons to the power output of the microwave;</u> or lost in microwave produced fed into cavity																			
(b)(iv)1.	 Correct direction of F_E drawn and labelled. (// to electric field line towards anode, see Fig. 7.3 for the electric field between the anode and cathode.) Correct direction of F_B drawn and labelled. (The velocity of the electron should be tangent to trajectory. Apply Fleming's left hand rule to get direction, towards centre of "circular motion" and perpendicular to trajectory) [-1] if F_E and F_B is not defined. Ignore weight if labelled. (Weight is insignificant to the electrical and magnetic forces)	1 1																		
(b)(iv)2.	As the <u>magnetic force on the electron is acting perpendicular to its motion</u>, (no work is done by the magnetic force on the electron), hence it does not affect the speed of motion as it moves towards the anode. Hence the <u>maximum kinetic gained remains unchanged</u>.	1 1																		
(c)(i)	<table border="1"> <thead> <tr> <th>depth into food z / mm</th><th>intensity of microwaves at depth z $I / \text{A.U.}$</th><th>$\ln(I / \text{A.U.})$</th></tr> </thead> <tbody> <tr> <td>0</td><td>24</td><td>3.18</td></tr> <tr> <td>4</td><td>19</td><td>2.94</td></tr> <tr> <td>8</td><td>15</td><td>2.71</td></tr> <tr> <td>12</td><td>12</td><td>2.49</td></tr> <tr> <td>16</td><td>10</td><td>2.30</td></tr> </tbody> </table> Values computed correctly and should follow the same d.p and s.f. of the relevant column	depth into food z / mm	intensity of microwaves at depth z $I / \text{A.U.}$	$\ln(I / \text{A.U.})$	0	24	3.18	4	19	2.94	8	15	2.71	12	12	2.49	16	10	2.30	1
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(c)(ii) 1. 2.	 1. Plot the point accurately to the nearest $\frac{1}{2}$ smallest square. 2. Appropriate best fit line drawn.	1 1
(c)(iii)	Correct computation of the gradient with two points on the best fit line. [-1] if the points are not read to $\frac{1}{2}$ square precision, $\frac{1}{2}$ square off . If more than one square off [0] for whole part. [-1] if the two points chosen are $\Delta z \leq 8$ mm (4 large squares) No units are required. However if units are incorrect [-1] E.g. Using (0, 3.16) and (16, 2.28), $\text{Gradient} = \frac{3.16 - 2.28}{0 - 16} = -0.055$	1 1
(c)(iv)	$I = I_0 e^{-\mu z} \Rightarrow \ln I = \ln I_0 - \mu z$ Hence, gradient of the graph = $-\mu$ When $z = \delta_p$, $I = I_0/e$ or $\ln(I/I_0) = 1/e$ or $I = 24/e$ $\Rightarrow \frac{I_0}{e} = I_0 e^{-\mu \delta_p} \Rightarrow \delta_p = 1/\mu = 1/0.055 = 18.1$ mm	1 1 1
(c)(v)1.	The potato mash with a higher water content would have <u>more of the microwave (energy) absorbed at the surface/ more microwave absorbed per unit length/per unit volume by the water molecules</u> the microwave moves through the food, (the intensity of the microwaves will then fall to 1/e of its intensity at the surface in a shorter depth). causing the penetration depth to be <u>smaller</u>.	1 1
(c)(v)2.	N – straight line graph with y-intercept unchanged, gradient is steeper. graph must be coherent with conclusion of penetration depth obtained earlier. [This mark is allocated on the page of the graph grids]	1

		Max mark	21
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